

Lecture 5. Dynamic Programming

Dynamic Programming for Discrete Time Optimal Control

- Given a state space $\mathbf{X}$, a control space $\mathbf{U}$, an observation space $\mathbf{Y}$, an initial state, noise models in stochastic case
- We need to design a control law that minimizes the objective function

	Deterministic Model	Stochastic Model
Continuous time	$\dot{x} = f(x, u)$ $y = h(x, u)$ $J = \int_{t=0}^T \delta^t l_t(x, u, \dot{x}) dt$	$\dot{x} = f(x, u, w)$ $y = h(x, u, v)$ $J = \int_{t=0}^T \delta^t l_t(x, u, \dot{x}) dt$
Discrete time	$x_{k+1} = f(x_k, u_k)$ $y_k = h(x_k, u_k)$ $J = \sum_{k=0}^N \delta^k l_k(x_k, u_k, x_{k+1})$	$x_{k+1} = f(x_k, u_k, w_k)$ $y_k = h(x_k, u_k, v_k)$ $J = \sum_{k=0}^N \delta^k l_k(x_k, u_k, x_{k+1})$

Problem Formulation

- Deterministic discrete time

- State feedback

$$\min_{u_0, u_1, \dots, u_{N-1}} J = \sum_{k=0}^{N-1} l_k(x_k, u_k) + l_N(x_N)$$

- Finite horizon

$$x_{k+1} = f(x_k, u_k)$$

- No discount

- Controllable

Problem Formulation

- Decision variables: the control sequences $\{u_0, \dots, u_{N-1}\}$

- Known parameters:

$$\min_{u_0, u_1, \dots, u_{N-1}} J = \sum_{k=0}^{N-1} l_k(x_k, u_k) + l_N(x_N)$$

- Initial state: x_0

$$x_{k+1} = f(x_k, u_k)$$

- Run-time costs: l_k

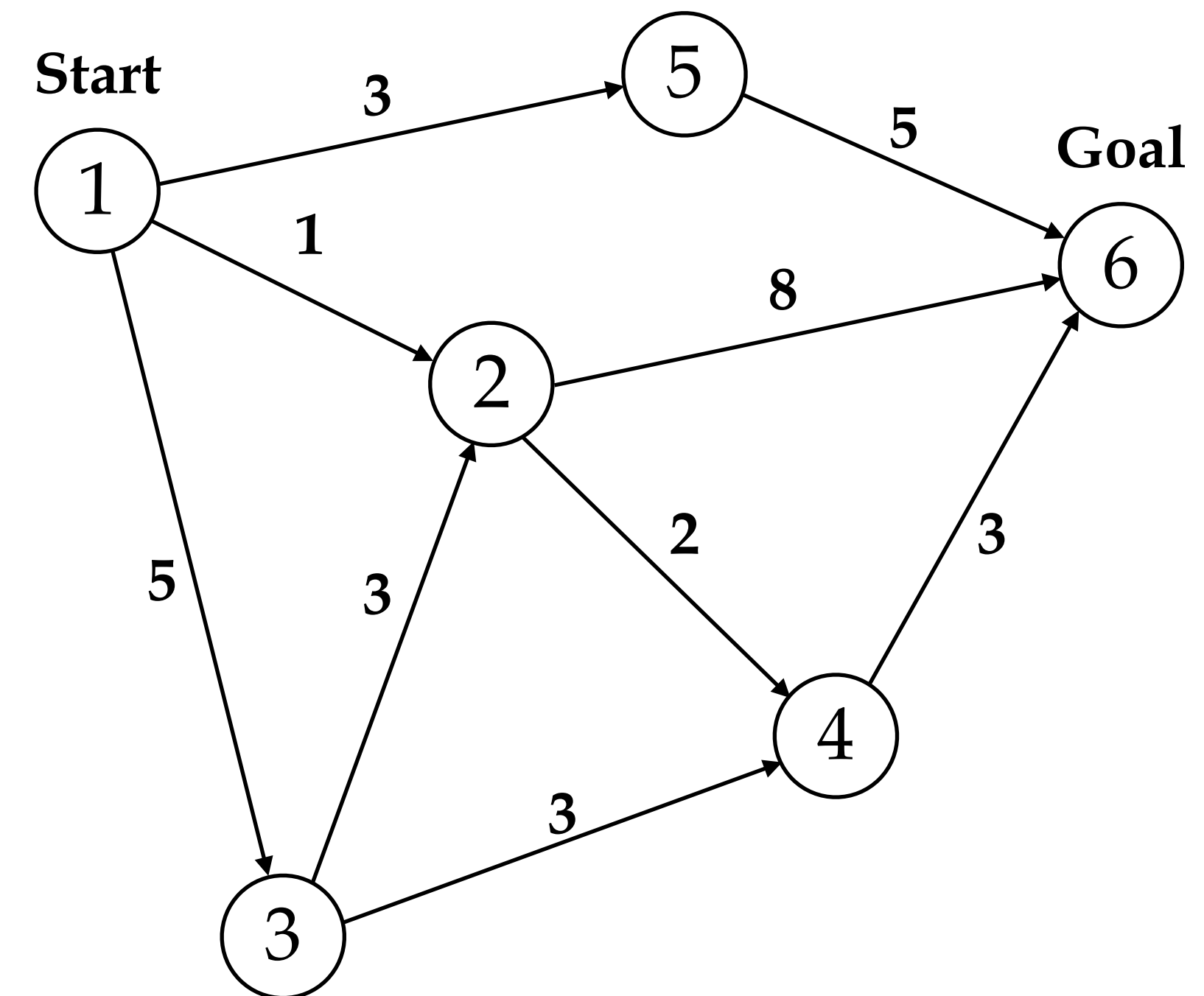
- Terminal cost: l_N

Solutions

- Open-loop solution:
 - A sequence of future controls $\{u_0, \dots, u_{N-1}\}$
- Closed-loop solution:
 - A control policy $u_k = q_k(x_k), \forall k$
 - More robust to disturbances

Example: Shortest Path Problem

- State space: nodes
- Action space: arcs
- Cost: distance traveled
- Termination: event triggered



Example: Regulation Problem

- Dynamic model (double integrator)

$$x_{k+1} = Ax_k + Bu_k$$

- Cost (quadratic)

$$J = \sum_{k=0}^{N-1} [x_k^T Q_k x_k + u_k^T R_k u_k] + x_N^T S_N x_N$$



- This is a linear quadratic regulation problem

Optimal Solution

$$\min_{u_0, u_1, \dots, u_{N-1}} J = \sum_{k=0}^{N-1} l_k(x_k, u_k) + l_N(x_N) \rightarrow J_0(x_0; u_0, \dots, u_{N-1})$$

$$x_{k+1} = f(x_k, u_k)$$

***We will not explicitly write this equality constraint if it is clear from the context.**

- Let $\{u_0^*, \dots, u_{N-1}^*\}$ be an optimal control sequence
- Together with x_0 , the optimal control sequence determines the corresponding state sequence $\{x_1^*, \dots, x_N^*\}$
- The optimal cost corresponding to the optimal sequence is denoted $J^*(x_0)$

Principle of Optimality

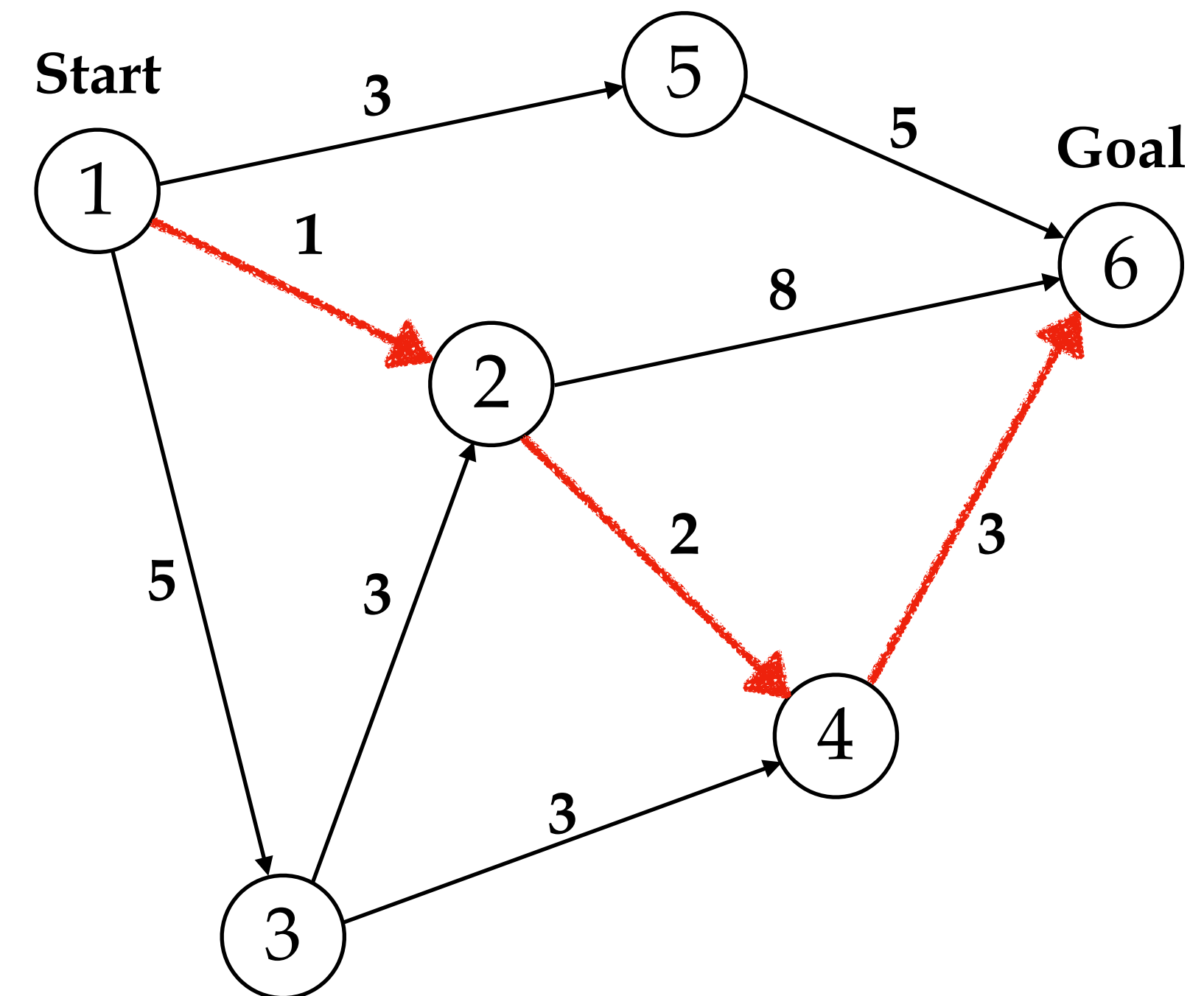
- Consider a subproblem: we start from x_k^* at time k and wish to minimize the “cost-to-go” from time k to time N over $\{u_k, \dots, u_{N-1}\}$,

$$\min_{u_k, \dots, u_{N-1}} l_k(x_k^*, u_k) + \sum_{m=k+1}^{N-1} l_m(x_m, u_m) + l_N(x_N) \longrightarrow J_k(x_k^*; u_k, \dots, u_{N-1})$$

- Assertion: the truncated optimal control sequence $\{u_k^*, \dots, u_{N-1}^*\}$ is optimal for this subproblem.

Example: Shortest Path Problem

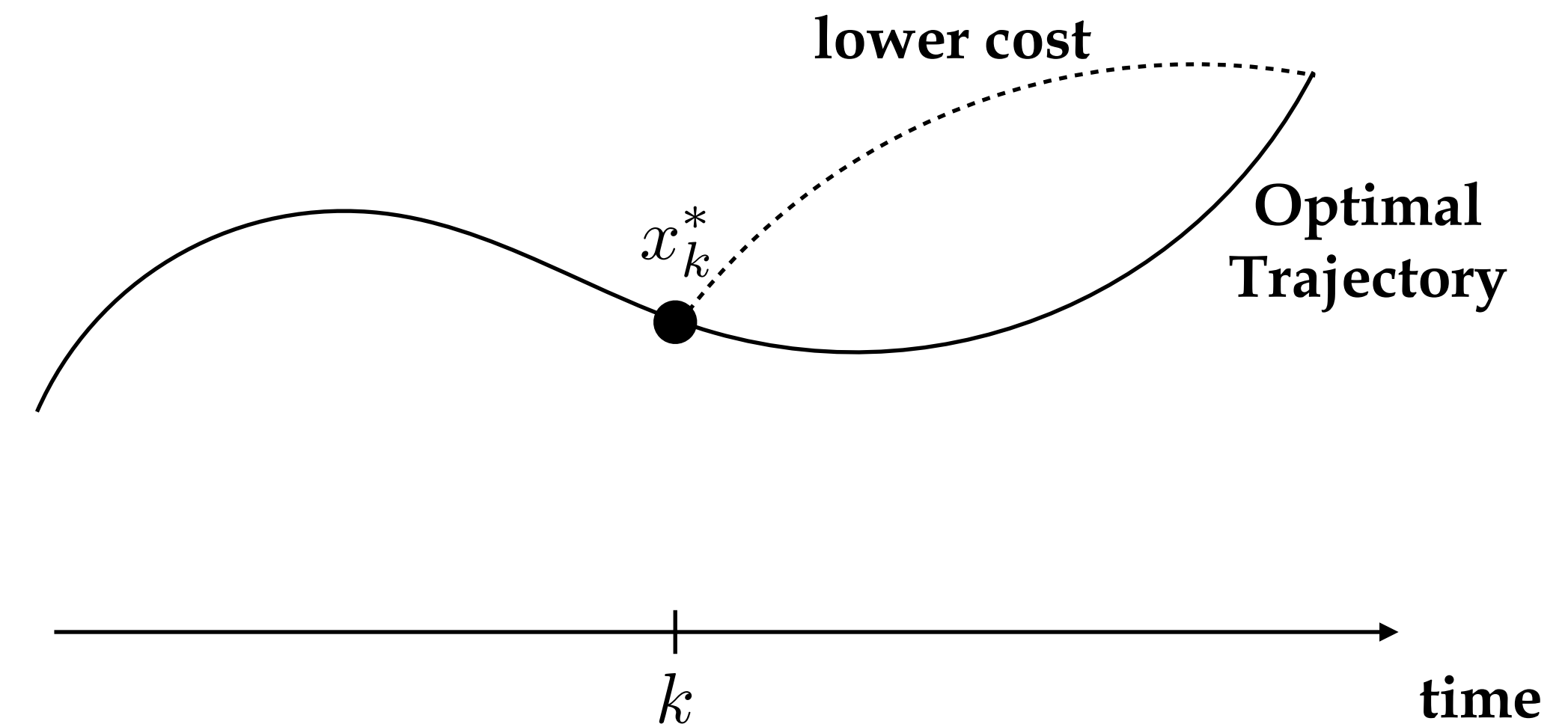
- 1-2-4-6 is the optimal path
- 2-4-6 is the optimal path from 2 to 6



Proof by Contradiction

- Suppose the truncated control sequence $\{u_k^*, \dots, u_{N-1}^*\}$ is not optimal for the subproblem.
- There exists another control sequence $\{u_k^o, \dots, u_{N-1}^o\}$ that will result in lower cost of the subproblem.

$$J_k(x_k^*; u_k^*, \dots, u_{N-1}^*) > J_k(x_k^*; u_k^o, \dots, u_{N-1}^o)$$



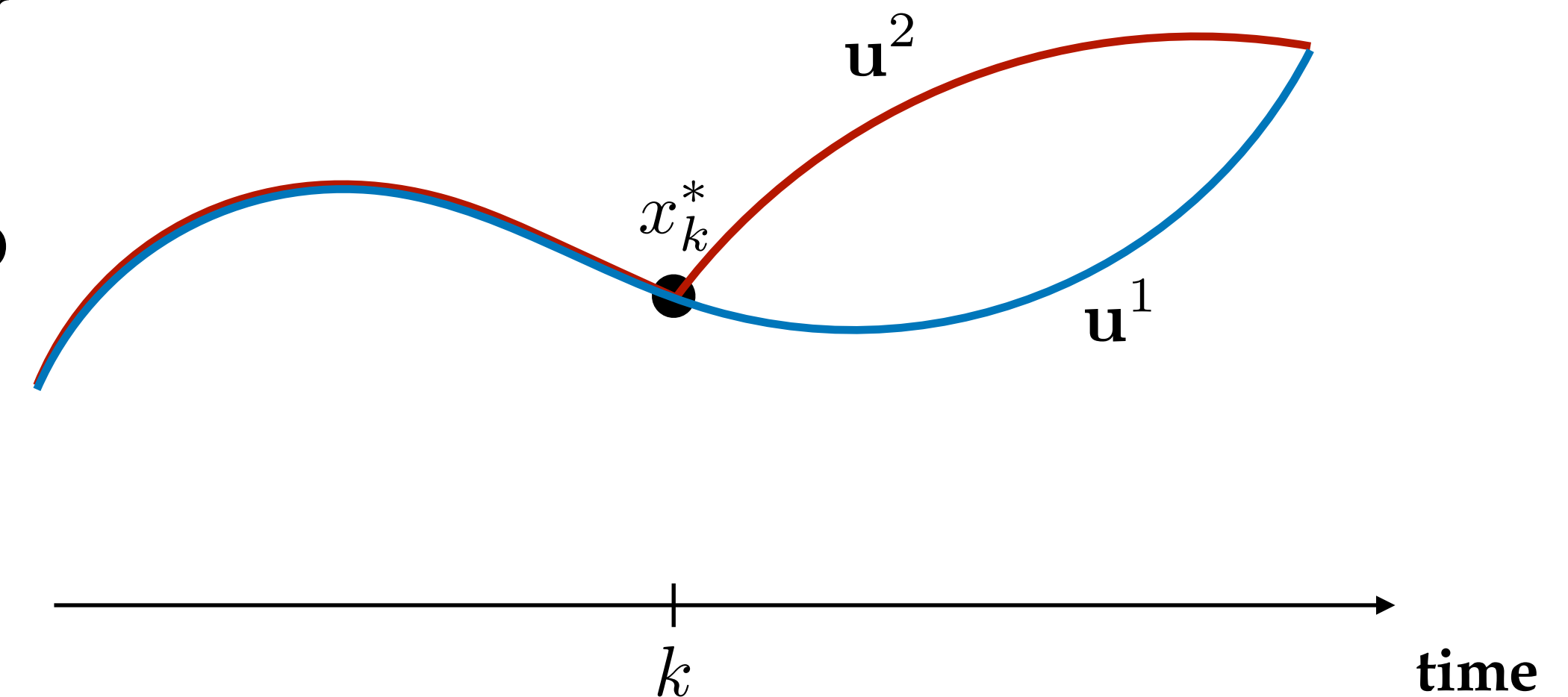
Proof by Contradiction

- Combining the optimal control sequence before k and the new control sequence starting at k : $\{u_0^*, \dots, u_{k-1}^*, u_k^o, \dots, u_{N-1}^o\}$

- Let's compare the total costs with respect to

- $\mathbf{u}^2 = \{u_0^*, \dots, u_{k-1}^*, u_k^o, \dots, u_{N-1}^o\}$

- $\mathbf{u}^1 = \{u_0^*, \dots, u_{N-1}^*\}$



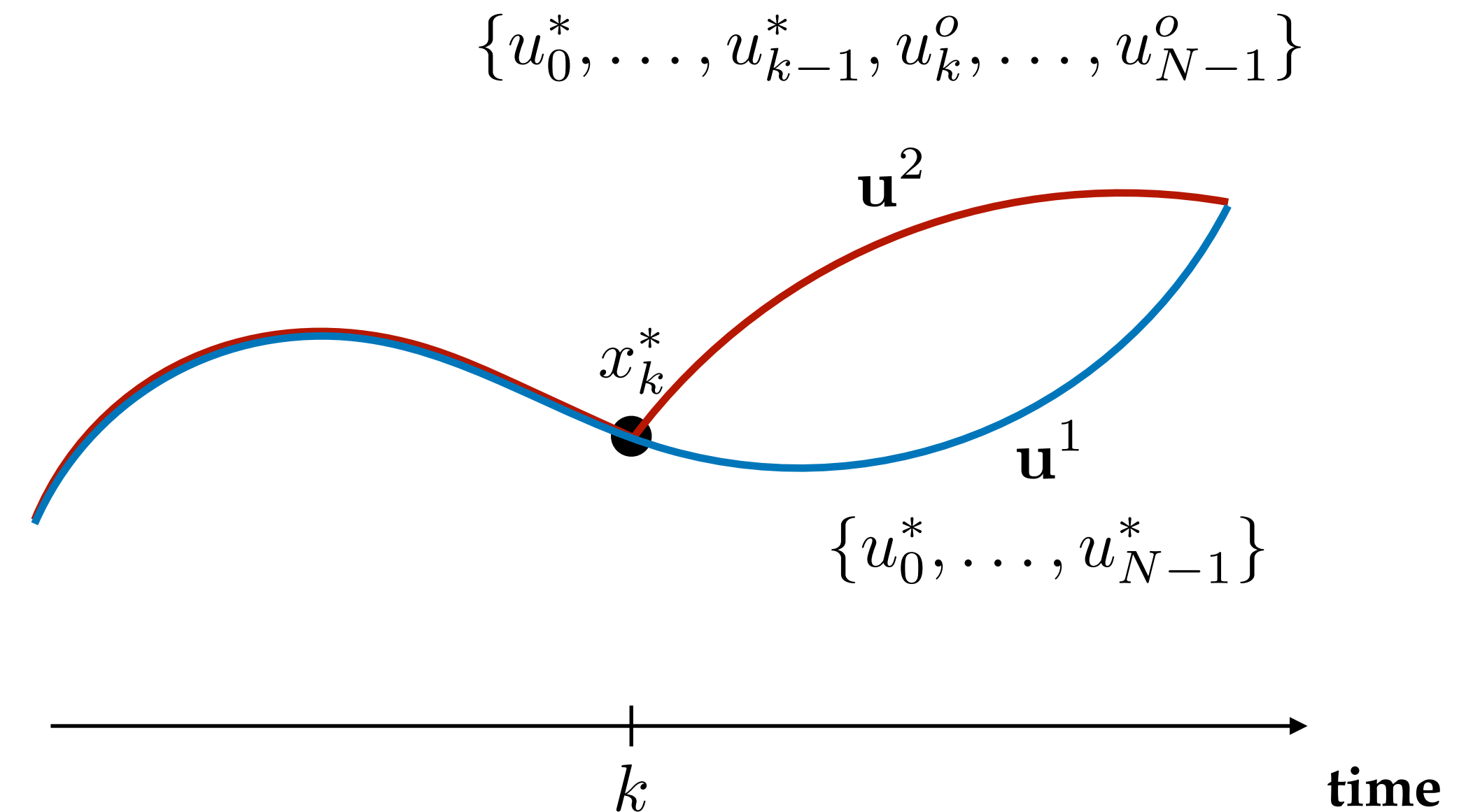
Proof by Contradiction

$$J_0(x_0; u_0, \dots, u_{N-1}) = \sum_{m=0}^{k-1} l_m(x_m, u_m) + J_k(x_k; u_k, \dots, u_{N-1})$$

$$J_0(x_0; \mathbf{u}^1) = \sum_{m=0}^{k-1} l_m(x_m^*, u_m^*) + J_k(x_k^*; u_k^*, \dots, u_{N-1}^*)$$

$$> \sum_{m=0}^{k-1} l_m(x_m^*, u_m^*) + J_k(x_k^*; u_k^o, \dots, u_{N-1}^o)$$

$$= J_0(x_0; \mathbf{u}^2)$$



Contradicts with the fact that $\mathbf{u}^1$ is an optimal control sequence!

Principle of Optimality

- Also known as Bellman Principle
- In plain word:
 - Any segment of an optimal trajectory is still optimal
 - An optimal trajectory won't have any suboptimal components
- The inverse is not true:
 - The concatenation of optimal trajectories may not be optimal

Utilizing Principle of Optimality

$$J_0(x_0; u_0, \dots, u_{N-1}) = \sum_{m=0}^{k-1} l_m(x_m, u_m) + J_k(x_k; u_k, \dots, u_{N-1})$$

Take minimization on both sides

$$\min_{u_0, \dots, u_{N-1}} J_0(x_0; u_0, \dots, u_{N-1}) = \min_{u_0, \dots, u_{k-1}} \left\{ \sum_{m=0}^{k-1} l_m(x_m, u_m) + \min_{u_k, \dots, u_{N-1}} J_k(\boxed{x_k}; u_k, \dots, u_{N-1}) \right\}$$


This term depends on $\{u_0, \dots, u_{k-1}\}$

These are not two independent minimization problems

But they can be solved sequentially

Utilizing Principle of Optimality

$$\min_{u_0, \dots, u_{N-1}} J_0(x_0; u_0, \dots, u_{N-1}) = \min_{u_0, \dots, u_{k-1}} \left\{ \sum_{m=0}^{k-1} l_m(x_m, u_m) + \min_{u_k, \dots, u_{N-1}} J_k(x_k; u_k, \dots, u_{N-1}) \right\}$$

$J_0^*(x_0)$
 $J_k^*(x_k)$ optimal cost-to-go

Let $k = 1$

$$J_0^*(x_0) = \min_{u_0} \{l_0(x_0, u_0) + J_1^*(x_1)\}$$

By induction

$$J_k^*(x_k) = \min_{u_k} \{l_k(x_k, u_k) + J_{k+1}^*(x_{k+1})\} \quad \forall k = 0, 1, \dots, N-1$$

$$J_N^*(x_N) = l_N(x_N)$$

Cost-to-Go and Value Function

Cost-to-go at time k: a function that depends on the current state and all future control inputs

$$J_k(x_k; u_k, \dots, u_{N-1}) = \sum_{m=k}^{N-1} l_m(x_m, u_m) + l_N(x_N)$$

Value function (optimal cost-to-go) at time k: a function that only depends on the current state (assuming optimal actions will be taken in the future)

$$J_k^*(x_k) = \min_{u_k, \dots, u_{N-1}} J_k(x_k; u_k, \dots, u_{N-1})$$

Bellman Equation

$$J_k^*(x_k) = \min_{u_k} \{l_k(x_k, u_k) + J_{k+1}^*(x_{k+1})\}$$

- Bellman equation is an update rule for value functions under the principle of optimality.
- General form:
 - Optimize over action
 - Optimization objective: run-time cost plus the value at next state
- We will see many variations of the Bellman equations in different scenarios.

Dynamic Programming

- Solving the optimal control sequence recursively using the principle of optimality
- At time N , we have

$$J_N^*(x_N) = l_N(x_N)$$

- For $k = N - 1, N - 2, \dots, 1, 0$, we solve the following optimization

$$J_k^*(x_k) = \min_{u_k} \{l_k(x_k, u_k) + J_{k+1}^*(x_{k+1})\}$$

and obtain the optimal control policy

$$q_k^*(x_k) = \arg \min_{u_k} \{l_k(x_k, u_k) + J_{k+1}^*(x_{k+1})\}$$

Dynamic Programming

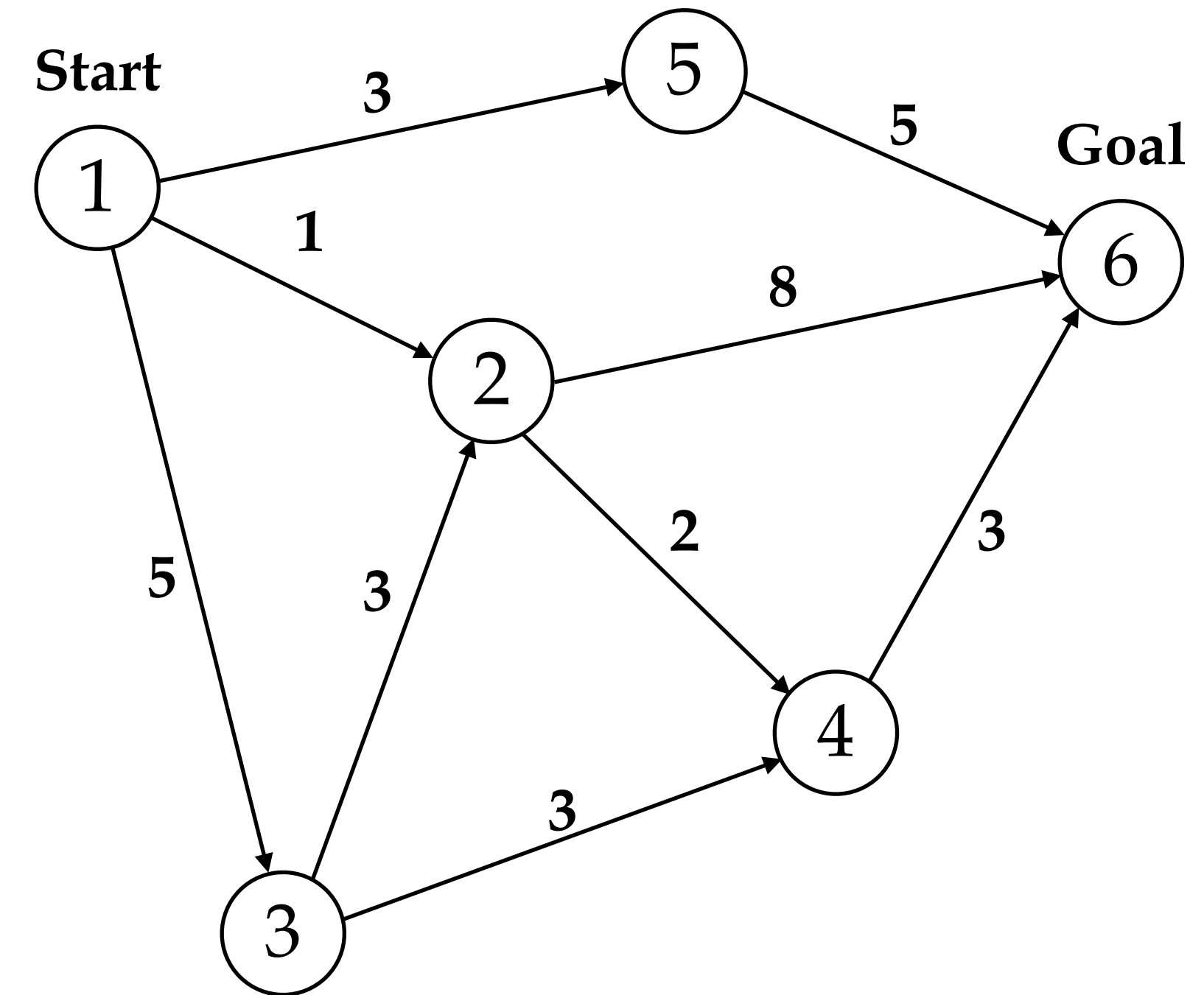
- Through this backward pass, we can only obtain the control policy as a mapping from the state to the control.
- The optimal control sequence needs to be obtained in a forward pass considering the initial state x_0

$$\begin{aligned}x_0^* &= x_0 \\ u_k^* &= q_k^*(x_k^*), \forall k = 0, 1, \dots, N-1 \\ x_{k+1}^* &= f(x_k^*, u_k^*), \forall k = 0, 1, \dots, N-1\end{aligned}$$

Example: Shortest Path Problem

	Optimal cost-to-go	Optimal action
Backward pass ↓	$J^*(6) = 0$	
	$J^*(5) = 5 + J^*(6) = 5$	5->6
	$J^*(4) = 3 + J^*(6) = 3$	4->6
	$J^*(2) = \min \{8 + J^*(6), \mathbf{2} + J^*(4)\} = 5$	2->4
	$J^*(3) = \min \{3 + J^*(2), \mathbf{3} + J^*(4)\} = 6$	3->4
	$J^*(1) = \min \{\mathbf{1} + J^*(2), 5 + J^*(3)\} = 6$	1->2

Forward pass
↗



Dynamic Programming for Continuous Time Optimal Control

- Given a state space $\mathbf{X}$, a control space $\mathbf{U}$, an observation space $\mathbf{Y}$, an initial state, noise models in stochastic case
- We need to design a control law that minimizes the objective function

	Deterministic Model	Stochastic Model
Continuous time	$\dot{x} = f(x, u)$ $y = h(x, u)$ $J = \int_{t=0}^T \delta^t l_t(x, u, \dot{x}) dt$	$\dot{x} = f(x, u, w)$ $y = h(x, u, v)$ $J = \int_{t=0}^T \delta^t l_t(x, u, \dot{x}) dt$
Discrete time	$x_{k+1} = f(x_k, u_k)$ $y_k = h(x_k, u_k)$ $J = \sum_{k=0}^N \delta^k l_k(x_k, u_k, x_{k+1})$	$x_{k+1} = f(x_k, u_k, w_k)$ $y_k = h(x_k, u_k, v_k)$ $J = \sum_{k=0}^N \delta^k l_k(x_k, u_k, x_{k+1})$

Problem Formulation

- Deterministic continuous time
- State feedback
- Finite horizon
- No discount
- Controllable System

$$\min_{u_t, t \in [0, T)} J = \int_{t=0}^T l_t(x_t, u_t) dt + l_T(x_T)$$

$$\dot{x} = f(x, u)$$

Problem Formulation

- Decision variables: the control trajectory $u_t, t \in [0, T)$

- Known parameters:

$$\min_{u_t, t \in [0, T)} J = \int_{t=0}^T l_t(x_t, u_t) dt + l_T(x_T)$$

- Initial state: x_0

$$\dot{x} = f(x, u)$$

- Run-time costs: l_t

- Terminal cost: l_T

Solutions

- Open-loop solution:
 - A trajectory of future controls $u_t, t \in [0, T)$
- Closed-loop solution:
 - A control policy $u_t = q_t(x_t), \forall t$
 - More robust to disturbances

Example: Regulation Problem

- Dynamic model (double integrator)

$$\dot{x} = Ax + Bu$$

- Cost (quadratic)

$$J = \int_{t=0}^T [x_t^T Q_t x_t + u_t^T R_t u_t] dt + x_T^T S_T x_T$$



- This is a linear quadratic regulation problem

Optimal Solution

$$\min_{u_t, t \in [0, T)} J = \int_{t=0}^T l_t(x_t, u_t) dt + l_T(x_T) \longrightarrow J_0(x_0; u_{0:T})$$

$$\dot{x} = f(x, u)$$

***We will not explicit write this equality constraint if it is clear from the context.**

- Let $u_{0:T}^*$ be an optimal control trajectory
- Together with x_0 , the optimal control sequence determines the corresponding state sequence $x_{0:T}^*$
- The optimal cost corresponding to the optimal sequence is denoted $J^*(x_0)$

Principle of Optimality

- Consider a subproblem: we start at x_t^* at time t and wish to minimize the “cost-to-go” from time t to time T over $u_{t:T}$,

$$\min_{u_{t:T}} \int_t^T l_\tau(x_\tau, u_\tau) d\tau + l_T(x_T)$$

- Assertion: the truncated optimal control sequence $u_{t:T}^*$ is optimal for this subproblem.
- We can also use proof by contradiction as discussed in the discrete time case.

Utilizing Principle of Optimality

$$J_t(x_t; u_{t:T}) = \int_t^{t'} l_\tau(x_\tau, u_\tau) d\tau + J_{t'}(x_{t'}; u_{t':T})$$

Take minimization on both sides

$$\min_{u_{t:T}} J_t(x_t; u_{t:T}) = \min_{u_{t:t'}} \left\{ \int_t^{t'} l_\tau(x_\tau, u_\tau) d\tau + \min_{u_{t':T}} J_{t'}(\boxed{x_{t'}}; u_{t':T}) \right\}$$



This term depends on $u_{t:t'}$

These are not two independent minimization problems

But they can be solved sequentially

Cost-to-Go and Value Function

Cost-to-go at time t : a function that depends on the current state and all future control inputs

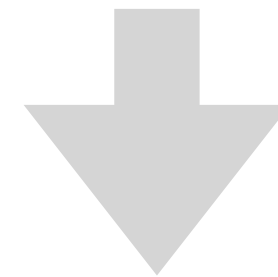
$$J_t(x_t; u_{t:T}) = \int_{\tau=t}^T l_{\tau}(x_{\tau}, u_{\tau}) d\tau + l_T(x_T)$$

Value function (optimal cost-to-go): a time-varying function that depends on the current state (assuming optimal actions will be taken in the future)

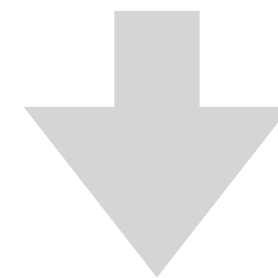
$$V(t, x_t) = \min_{u_{t:T}} J_t(x_t; u_{t:T})$$

Utilizing Principle of Optimality

$$\underbrace{\min_{u_{t:T}} J_t(x_t; u_{t:T})}_{V(t, x_t)} = \min_{u_{t:t'}} \left\{ \int_t^{t'} l_\tau(x_\tau, u_\tau) d\tau + \underbrace{\min_{u_{t':T}} J_{t'}(x_{t'}; u_{t':T})}_{V(t', x_{t'}) \text{ optimal cost-to-go}} \right\}$$



$$0 = \min_{u_{t:t'}} \left\{ \int_t^{t'} l_\tau(x_\tau, u_\tau) d\tau + V(t', x_{t'}) - V(t, x_t) \right\}$$



$$0 = \lim_{t' \rightarrow t^+} \min_{u_{t:t'}} \left\{ \frac{1}{t' - t} \int_t^{t'} l_\tau(x_\tau, u_\tau) d\tau + \frac{V(t', x_{t'}) - V(t, x_t)}{t' - t} \right\}$$

Utilizing Principle of Optimality

$$0 = \lim_{t' \rightarrow t^+} \min_{u_{t:t'}} \left\{ \frac{1}{t' - t} \int_t^{t'} l_\tau(x_\tau, u_\tau) d\tau + \frac{V(t', x_{t'}) - V(t, x_t)}{t' - t} \right\}$$

Under smoothness assumptions of all functions

$$\lim_{t' \rightarrow t^+} \frac{1}{t' - t} \int_t^{t'} l_\tau(x_\tau, u_\tau) d\tau = l_t(x_t, u_t)$$

$$\lim_{t' \rightarrow t^+} \frac{V(t', x_{t'}) - V(t, x_t)}{t' - t} = \frac{\partial}{\partial t} V(t, x_t) + \frac{\partial}{\partial x} V(t, x_t) \cdot \dot{x}_t$$

Hamilton-Jacobi-Bellman equation

$$0 = \frac{\partial}{\partial t} V(t, x_t) + \min_{u_t} \{ l_t(x_t, u_t) + \frac{\partial}{\partial x} V(t, x_t) \cdot f(x_t, u_t) \}$$

Hamilton-Jacobi-Bellman Equation

$$0 = \underbrace{\frac{\partial}{\partial t} V(t, x_t)}_{\dot{V}(t, x_t)} + \min_{u_t} \{ l_t(x_t, u_t) + \underbrace{\frac{\partial}{\partial x} V(t, x_t) \cdot f(x_t, u_t)}_{\nabla_x V(t, x_t)} \}$$

- HJB equation is a nonlinear partial differential equation regarding the value function.
- Its discrete-time version is the Bellman equation.

$$J_k^*(x_k) = \min_{u_k} \{ l_k(x_k, u_k) + J_{k+1}^*(x_{k+1}) \}$$

Hamilton-Jacobi-Bellman Equation



William Hamilton



Carl Jacobi



Richard Bellman

Dynamic Programming

- Solving the optimal control recursively by solving the HJB equation
- At time T , we have

$$V(T, x_T) = l_T(x_T)$$

- We solve the following optimization backward in time

$$\dot{V}(t, x_t) = - \min_{u_t} \{ l_t(x_t, u_t) + \nabla_x V(t, x_t) \cdot f(x_t, u_t) \}$$

and obtain the optimal control policy

$$q_t^*(x_t) = \arg \min_{u_t} \{ l_t(x_t, u_t) + \nabla_x V(t, x_t) \cdot f(x_t, u_t) \}$$

Dynamic Programming

- Through this backward pass, we can only obtain the control policy as a mapping from the state to the control.
- The optimal control sequence needs to be obtained in a forward pass considering the initial state x_0

$$x_0^* = x_0$$

$$u_t^* = q_t^*(x_t^*), \forall t \in [0, T)$$

$$\dot{x}_t^* = f(x_t^*, u_t^*), \forall t \in [0, T]$$

Exercise

- Dynamics:

$$\dot{x} = u, |u| \leq 1, t \in [0, T]$$

- Cost function

$$J(x_0) = \min_u |x(T)|$$

- What is the HJB equation? What is the optimal control strategy?

Next Time

- Maximum Principle